

DILLINGHAM, AK, USA

WMO: 703210

Lat: 59.050N

Lon: 158.517W

Elev: 26

StdP: 101.01

Time Zone: -9.00 (NAK)

Period: 94-19

WBAN: 25513

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-27.7	-25.0	-32.7	0.2	-26.2	-30.9	0.2	-23.7	14.2	-6.8	12.4	-5.8	1.5	350	0.760

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	6.7	21.5	14.9	19.3	14.1	17.8	13.1	15.9	19.9	14.8	18.4	13.9	16.9	4.0	340

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.0	10.0	16.9	13.0	9.4	15.7	12.4	9.0	15.1	44.5	20.2	41.3	18.4	38.8	16.8	22.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.2	9.7	8.5	DB	-30.2	24.5	4.1	2.6	-33.2	26.4	-35.5	27.9	-37.8	29.4	-40.8	31.3	(4)
(5)			WB	-29.8	17.4	3.6	2.1	-32.4	18.9	-34.5	20.1	-36.5	21.3	-39.1	22.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	1.8	-8.9	-6.3	-5.9	0.6	6.5	11.0	13.0	12.5	8.9	2.4	-4.6	-7.7	(6)	
	DBStd	9.64	8.40	7.35	6.29	4.40	3.45	2.73	2.57	2.46	2.71	4.53	6.38	7.94	(7)	
	HDD10.0	3231	586	455	492	283	118	18	3	6	49	236	437	548	(8)	
	HDD18.3	6024	844	688	750	533	367	221	167	182	283	494	687	806	(9)	
	CDD10.0	251	0	0	0	0	9	47	96	82	17	0	0	0	(10)	
	CDD18.3	2	0	0	0	0	0	1	2	0	0	0	0	0	(11)	
	CDH23.3	23	0	0	0	0	1	7	13	2	0	0	0	0	(12)	
	CDH26.7	3	0	0	0	0	0	1	2	0	0	0	0	0	(13)	
(14) Wind	WSAvg	3.8	4.6	4.5	4.1	4.0	3.7	3.2	3.1	3.1	3.4	3.8	4.0	4.2	(14)	
(15) Precipitation	PrecAvg	843	70	58	53	42	49	57	74	82	101	86	87	82	(15)	
	PrecMax	1220	164	116	154	78	106	93	123	133	202	158	196	184	(16)	
	PrecMin	540	16	10	4	7	7	28	37	34	21	8	12	17	(17)	
	PrecStd	154	32	28	36	19	23	15	21	27	39	33	42	35	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.9	4.0	6.1	12.5	21.0	23.8	25.0	22.4	17.4	12.1	7.7	4.9	(19)	
		MCWB	2.4	2.4	3.2	6.5	11.4	14.9	17.4	16.0	11.9	9.0	5.7	2.9	(20)	
	2%	DB	2.9	2.9	3.2	9.9	17.3	20.8	22.0	20.0	15.5	10.5	6.4	3.8	(21)	
		MCWB	1.5	1.6	1.0	5.4	9.9	13.5	15.8	15.1	11.6	8.4	4.9	2.2	(22)	
	5%	DB	2.4	2.4	2.5	7.7	14.3	18.1	19.6	18.0	13.9	9.1	4.9	2.7	(23)	
		MCWB	1.1	1.2	0.7	4.2	8.5	12.9	14.9	13.7	10.7	7.5	3.6	1.4	(24)	
	10%	DB	1.5	2.0	2.0	6.2	12.4	16.6	17.8	17.0	12.8	8.0	3.1	2.1	(25)	
		MCWB	0.5	0.8	0.5	3.3	7.7	11.8	13.6	13.1	10.2	6.5	1.9	1.1	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.0	3.1	3.6	7.4	12.4	16.0	18.9	17.1	14.1	10.0	6.6	3.4	(27)	
		MCDB	3.6	3.5	5.4	11.4	18.8	21.2	23.1	20.8	15.8	11.0	7.1	4.6	(28)	
	2%	WB	2.1	2.2	1.8	5.6	10.5	14.5	16.4	15.6	12.6	8.9	5.2	2.5	(29)	
		MCDB	2.8	2.8	2.8	9.0	16.1	19.2	20.7	18.8	14.1	9.9	6.7	3.5	(30)	
	5%	WB	1.2	1.3	1.1	4.6	9.2	13.4	15.1	14.5	11.7	7.9	3.9	1.7	(31)	
		MCDB	2.1	2.1	2.1	7.5	13.6	17.5	19.1	17.2	13.1	8.8	4.9	2.6	(32)	
	10%	WB	0.4	0.6	0.4	3.6	8.1	12.2	14.1	13.7	10.7	6.9	2.4	0.9	(33)	
		MCDB	1.4	1.5	1.5	5.9	11.5	15.9	17.2	16.1	12.3	8.0	3.3	1.8	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	5.8	6.1	6.9	6.6	7.7	7.7	6.7	6.5	6.5	5.8	6.1	6.1	(35)	
		MCDBR	4.0	4.4	5.2	8.6	12.0	11.2	10.8	9.1	7.9	5.5	4.6	4.0	(36)	
	5% WB	MCWBR	3.5	4.0	4.1	5.5	6.9	6.3	6.0	5.4	5.0	4.1	4.1	3.8	(37)	
		MCWBR	3.9	4.4	4.7	8.0	10.6	10.1	9.6	7.8	6.3	4.6	4.5	4.1	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.259	0.299	0.312	0.355	0.393	0.361	0.364	0.369	0.332	0.307	0.289	0.259	(40)	
		taud	2.121	2.180	2.186	2.189	2.228	2.421	2.452	2.431	2.512	2.425	2.095	1.987	(41)	
	Ebn at Noon		543	708	820	843	834	873	862	828	809	713	475	371	(42)	
		Edh at Noon	47	76	101	120	125	105	100	95	75	60	48	38	(43)	
(44) All-Sky Solar Radiation	RadAvg		0.41	1.04	2.82	4.40	5.17	5.26	4.48	3.62	2.53	1.31	0.53	0.23	(44)	
	RadStd		0.08	0.20	0.44	0.54	0.73	0.46	0.47	0.44	0.33	0.16	0.08	0.03	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (7 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page